Math 1324 – Final Exam Review Problems

1. Solve: $3x - 17 = 5x + 5$
2. Solve: $2(4 - 3x) + 7 = 3(10 - x) - 3x$
3. For the functions $f(x) = x^2 + 2,\ g(x) = 3x - 7,\ h(x) = \frac{6}{x-1}$, find the following.
   (a) $g(8)$
   (b) $f(a + 3)$
   (c) $h\big(g(3)\big)$
   (d) $(f - g)(1)$
   (e) $\left(\frac{g}{h}\right)(-1)$
   (f) $(fg)(-2)$
   (g) $g\big(f(4)\big)$
4. Find the equation for each of the following lines.
   (a) The line contains the points $(-1, 4)$ and $(2, -2)$.
   (b) The line is parallel to $y = 3x - 4$ and contains the point $(2, 3)$.
   (c) The line is perpendicular to $y = \frac{2}{3}x + 7$ and contains the point $(4, -1)$.
5. Solve: $\begin{cases} 4x - y = 3 \\ 2x + 3y = 19 \end{cases}$
6. Solve: $\begin{cases} 2x - 3y = 5 \\ 6x - 9y = 12 \end{cases}$
7. Solve: $\begin{cases} x - 2y + z = 6 \\ 2x + y + z = 5 \\ x + 3y - 2z = -5 \end{cases}$
8. Find the exact solutions to $25x^2 - 9 = 0$.
9. Find the exact solutions to $w^2 - 8w + 15 = 0$.
10. Find the exact solutions to $2y^2 - y - 4 = 0$.
11. If the profit from the sale of $x$ units of a product is $P(x) = 16x - 0.1x^2 - 100$, what levels of production will yield a profit of \$180?
12. Find the vertex of $y = -3x^2 + 6x - 5$.
13. Determine the maximum/minimum value of $y = -10(x + 3)^2 + 9$.
14. Find the zeros of $y = x^2 + 10x - 7$.
15. A ball thrown vertically into the air has its height above the ground given by $s = 112t - 16t^2$ where $t$ is in seconds and $s$ is in feet. Find the maximum height of the ball.
16. Suppose the manufacturer of a board game will supply 10,000 games if the wholesale price is \$1.50 each but will supply only 5000 if the price is \$1.00 each. Assuming that the supply function is linear, write its equation.
17. Find the market equilibrium point if demand is $p = -2q + 320$ and supply is $p = 8q + 20$.
18. Suppose the supply function is $p = q^2 + 8q + 22$ and the demand is $p = 198 - 4q - \frac{1}{4}q^2$. Find the market equilibrium quantity and price, then determine the revenue generated at the market equilibrium.
19. Find the maximum revenue generated by the revenue function $R(x) = 1600x - x^2$.
20. If a company has total costs $C(x) = 15000 + 35x + 0.1x^2$ and total revenues $R(x) = 385x - 0.9x^2$, find the break-even points and the profit function.
21. The profit function for a firm making widgets is $P(x) = 88x - x^2 - 1200$. Find the number of units at which maximum profit is achieved, and find the maximum profit.

22. Consider the matrices $A = \begin{bmatrix} 3 & -2 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -2 \\ 4 & 5 \end{bmatrix}$. Find the following.

(a) $5A$
(b) $2A - 3B$
(c) $B^T$
(d) $AB$
(e) $A^T B$
(f) $B^T - A$

23. Write an augmented matrix for the system: $\begin{cases} x - 3y + z = 5 \\ y + 2z = 0 \\ -x + 4y + 2z = 7 \end{cases}$

24. What is the solution to the system with the following reduced row echelon form augmented matrix?

(a) $\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 3 \end{array}\right]$

(b) $\left[\begin{array}{ccc|c} 1 & -2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$

(c) $\left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$

25. Solve the following systems using matrices.

(a) $\begin{cases} 2x + y - z = 2 \\ x - y + 2z = 3 \\ x + y - z = 1 \end{cases}$

(b) $\begin{cases} x - 3y + z = -4 \\ 2x - 5y - 5z = -9 \end{cases}$

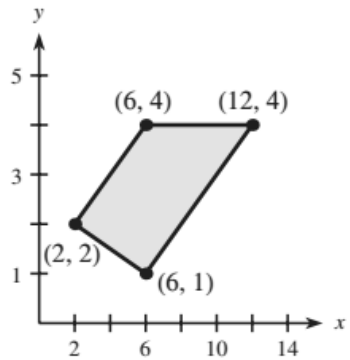
(c) $\begin{cases} x + 3y + 2z = 2 \\ 2x - y - 2z = 1 \\ 3x + 2y = 1 \end{cases}$

26. Graph the solution to $y \leq 2x + 4$.

27. Graph the solution to $3x - 4y < 12$.

28. Graph the solution to the system $\begin{cases} x + 5y \leq 200 \\ 2x + 3y \leq 134. \\ x \geq 0,\ y \geq 0 \end{cases}$

29. Find the maximum and minimum value of $f = 2x + 9y$ over the region below.




30. Maximize $f = x + 2y$ subject to

$x + y \geq 4$
$2x + y \leq 8$
$y \leq 4$
$x \geq 0, y \geq 0$

31. Use the simplex method to maximize $f = x + 3y + z$ subject to the constraints below. (Assume $x \geq 0, y \geq 0$.)

$$\begin{aligned} x + 4y \quad &\leq 12 \\ 3x + 6y + 4z &\leq 48 \\ y + \ z &\leq \ 8 \end{aligned}$$

32. Write the equation $2^7 = 128$ in logarithmic form.
33. Write the equation $\log_7(x + 3) = -1$ in exponential form, then solve for $x$.
34. Simplify the following logarithms.
    (a) $\log_3 27$
    (b) $\log_4 1$
35. Split into simple logarithms: $\ln\left(x^3\sqrt{x+1}\right)$
36. Combine into a single logarithm: $2\ln x + 3\ln y - \ln z$
37. Solve the following equations. Give approximate solutions.
    (a) $5^{x-3} = 17$
    (b) $3 - 2\log_3 x = 1$
38. The sales of a product decline after the end of an advertising campaign, with the sales decay given by $S = 100{,}000e^{-0.5t}$, where $S$ represents the weekly sales and $t$ represents the number of weeks since the end of the campaign.
    (a) What will be the sales for the tenth week after the end of the campaign?
    (b) During what week after the end of the campaign will sales drop below 400?
39. $2400 is invested for 9 months at an annual simple interest rate of 15%.
    (a) How much interest will be earned?
    (b) What is the future value of the investment after 3 months?
40. If $5000 is invested at 6% annual simple interest, how long does it take to be worth $9000?
41. If $2000 is invested with a 9% annual interest rate, find the following.
    (a) the value after 15 years if interest is compounded monthly
    (b) the value after 15 years if interest is compounded continuously
    (c) how long it will take to grow to $12000 if interest is compounded continuously
42. How much money must be invested now to have a value of $800 in 6 years if the annual interest rate is 5.2%, compounded quarterly?
43. $300 will be deposited in an annuity at the end of each month for 20 years and earns 6% interest, compounded monthly.
    (a) What will the value be in 20 years?
    (b) What is the present value of the annuity?
44. A state lottery prize of $4,000,000 will be paid out in payments of $400,000 at the beginning of each year for the next 10 years. The winner also has the option of taking the present value of this amount in one lump sum. If the state assumes a 4% interest rate, compounded annually, what would the lump sum be?
45. An inheritance of $30,000 is invested at 4.8% interest, compounded monthly. If $1000 payments are made from the account at the end of each month, how long will it take the account to decrease to $0?
46. A couple looking for a 30-year mortgage can get a 3.6% annual interest rate, compounded quarterly. They think they could afford a $2200 quarterly payment on the principle and interest for the loan, with payments due at the end of each quarter. How much money could they borrow?
47. How much must be deposited at the beginning of each month to have $100,000 in 7 years if the annual interest rate is 5.4%, compounded monthly?

48. Answer the following based on two fair six-sided dice being rolled.

| | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| 1 | (1, 1) | (1, 2) | (1, 3) | (1, 4) | (1, 5) | (1, 6) |
| 2 | (2, 1) | (2, 2) | (2, 3) | (2, 4) | (2, 5) | (2, 6) |
| 3 | (3, 1) | (3, 2) | (3, 3) | (3, 4) | (3, 5) | (3, 6) |
| 4 | (4, 1) | (4, 2) | (4, 3) | (4, 4) | (4, 5) | (4, 6) |
| 5 | (5, 1) | (5, 2) | (5, 3) | (5, 4) | (5, 5) | (5, 6) |
| 6 | (6, 1) | (6, 2) | (6, 3) | (6, 4) | (6, 5) | (6, 6) |

   (a) What is the probability that the sum is 5?
   (b) What is the probability that the sum is at least 10 and at least one die is odd?
   (c) What is the probability that the sum is 6 or at least one die is 4?
   (d) What is the probability that the sum is 7 given that at least one die is 5?

49. Answer the following based on two balls being drawn without replacement from a bag containing 6 green balls and 8 yellow balls.
   (a) What is the probability that the first ball is green?
   (b) What is the probability that both balls are green?
   (c) What is the probability that the second ball is yellow, given that the first ball is green?

50. Two urns contain balls. Urn I contains 6 blue balls and 4 red balls, while Urn II contains 2 blue balls and 4 red balls. An urn is selected at random and then a random ball is drawn. Answer the following.
   (a) What is the probability that the ball is blue?
   (b) What is the probability that the ball came from Urn I, given that the ball is red?

51. One hundred boys and 100 girls were asked whether they had ever been frightened by a television program. Thirty of the boys and 60 of the girls said they had been frightened. One of these children is selected at random.
   (a) What is the probability that he or she has been frightened?
   (b) What is the probability that child is a girl, given that he or she has been frightened?

52. Product testing shows that 20% of light bulbs of a certain brand die within a year of normal use. Three light bulbs are chosen at random.
   (a) What is the probability that all three will die within a year?
   (b) What is the probability that none of the light bulbs will die within a year?

53. Find the mean and standard deviation of the discrete random variable with the probability distribution below.

| $x$ | $-5$ | $-1$ | $4$ | $6$ |
|---|---|---|---|---|
| $\Pr(x)$ | $0.2$ | $0.3$ | $0.15$ | $0.35$ |